

Damped Permutation Scheduling for Coupled Flood–Seismicity Models: A Tractable Stability Remedy for Full Coupling

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Abstract

Multi-hazard early-warning systems that couple hydrological and seismic subsystems must reconcile two data streams that arrive asynchronously. A natural modelling device is to let a pre-recorded binary schedule decide, at each step, which coupling direction is active, with the exchange represented by a permutation matrix. We formalise this device, define an exact coupling coefficient, and show that the pure transposition produces a permuted field with a strictly positive eigenvalue and is therefore unstable for every step size. We then introduce a single *damping parameter* $\beta \in [0, 1/2)$ that convexly blends the natural and permuted increments, and prove that the damped full-coupling schedule has both eigenvalues of its active field strictly in the left half-plane for every $\beta < 1/2$, with a closed-form stability window that grows monotonically from the uncoupled bound to $4/(a + b)$ as $\beta \rightarrow 1/2^-$. The undamped paper is recovered at $\beta = 1$, where the window collapses to zero. A period-two alternating schedule recovers the same bound at $C_N = 1/2$, with monodromy available in closed form. A stylized flood–seismicity test bed and runnable code accompany the analysis. The practical conclusion is that coupling strength alone does not determine stability, and that both the full schedule and the exchange strength β must be

24 reported.

25 **Keywords:** multi-hazard coupling, deterministic scheduling, switched systems, reservoir-

26 induced seismicity, numerical stability, damped permutation

1 Introduction

Operational multi-hazard platforms increasingly fuse hydrological and seismic data into a single forecasting pipeline. Reservoir operations, for example, are monitored alongside local seismicity because filling and drawdown are known to modulate earthquake occurrence [9, 2], and conversely because ground shaking alters permeability, dam integrity, and gauge behaviour [8]. The two subsystems are physically coupled in both directions, but the coupling is rarely active in both directions at once. Data arrive on different clocks: water-level gauges report hourly, seismic networks report event-triggered, and the integration step is often chosen to match neither.

When the exchange pattern is dictated by a recorded event log rather than by a random process, the resulting integrator is deterministic but *state-dependent in structure*. The simplest formalisation is a binary schedule: at step k a value $s_k \in \{0, 1\}$ selects whether the vector field is evaluated in its natural ordering or in a permuted ordering. The permutation is fixed and represented by a permutation matrix $\mathbf{P}$, and we introduce a damping parameter β that controls the strength of the exchange.

This construction sits at the intersection of three literatures. It is a special case of operator splitting and composition [10, 7]; it is a switched or hybrid system in the sense of Liberzon [6]; and because the exchange operator is a convex combination of the identity and a permutation, it sits between geometric and non-geometric integrators [3, 5]. We show that the pure transposition ($\beta = 1$) is misleadingly similar to a geometric integrator, while the damped version ($\beta < 1/2$) is a well-posed, stabilizable scheme.

Our contributions are four. First, we give a self-contained formulation with an exact coupling coefficient and a strength parameter (Sections 2–3). Second, we show that the fourth-order Runge–Kutta extension retains its classical order when the switch is frozen across each step (Section 4). Third, we prove that for a two-block linearisation the pure transposition has a positive real eigenvalue and is unstable for every step size, while the damped operator with $\beta < 1/2$ has both eigenvalues strictly negative (Section 5). Fourth, we give closed-form stability windows for the damped full-coupling and alternating schedules, and demonstrate that the window grows monotonically from the uncoupled bound to $4/(a + b)$ as $\beta \rightarrow 1/2^-$. Section 6

sets out a stylized flood–seismicity test bed, Section 7 discusses scope, and Section 8 concludes.

2 Formulation

2.1 The damped scheduled explicit Euler map

Let the state at step k be $\mathbf{x}_k \in \mathbb{R}^n$, let $h > 0$ be the step size, let $\mathbf{P} \in \mathbb{R}^{n \times n}$ be a fixed permutation matrix, and let $\beta \in [0, 1]$ be a fixed *exchange strength* parameter. Let $\{s_k\}_{k \geq 0} \subset \{0, 1\}$ be a deterministic binary schedule supplied by the user. The damped scheduled explicit Euler iteration is

$$\mathbf{x}_{k+1} = \mathbf{x}_k + h \mathbf{F}^{(s_k)}(\mathbf{x}_k), \quad (1)$$

with the scheduled field

$$\mathbf{F}^{(s_k)}(\mathbf{x}_k) = \mathbf{L}_{s_k} \mathbf{F}(\mathbf{x}_k), \quad \mathbf{L}_s := \mathbf{I} + s \beta (\mathbf{P} - \mathbf{I}). \quad (2)$$

Expanding the operator gives the working form

$$\boxed{\mathbf{x}_{k+1} = \mathbf{x}_k + h \left((1 - s_k) \mathbf{F}(\mathbf{x}_k) + s_k [(1 - \beta) \mathbf{F}(\mathbf{x}_k) + \beta \mathbf{P} \mathbf{F}(\mathbf{x}_k)] \right)}. \quad (3)$$

Three remarks. When $\beta = 0$ the schedule has no effect for any s_k , so the construction is non-trivial only for $\beta \neq 0$ and $\mathbf{P} \neq \mathbf{I}$. When $\beta = 1$ the operator recovers the pure transposition, which we will show in Section 5 is unconditionally unstable. And because the schedule enters linearly, the map is still a convex combination of two fixed one-step maps, so the linearity used for the Runge–Kutta stages below is preserved.

2.2 Two-block structure

For the multi-hazard setting we take $n = 2$ with $\mathbf{x} = (u, v)^\top$, where u is a hydrological variable (reservoir stage or pore pressure) and v is a seismic variable (slip deficit or stress state). The

73 permutation

$$\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (4)$$

74 exchanges the two coupling directions. If $s_k = 0$ the field is evaluated in its natural ordering,

$$\begin{cases} u_{k+1} = u_k + h F_u(u_k, v_k), \\ v_{k+1} = v_k + h F_v(u_k, v_k), \end{cases} \quad (5)$$

75 and if $s_k = 1$ the increments are damped-exchange weighted,

$$\begin{cases} u_{k+1} = u_k + h[(1 - \beta)F_u(u_k, v_k) + \beta F_v(u_k, v_k)], \\ v_{k+1} = v_k + h[\beta F_u(u_k, v_k) + (1 - \beta)F_v(u_k, v_k)]. \end{cases} \quad (6)$$

76 The state remains in the same coordinate basis; only the increment is a β -weighted blend of the
77 natural and permuted forms.

78 2.3 Algorithm

79 For general n and general $\mathbf{P}$, the iteration is:

```
80 Input:  x_0                initial state, length n
81        F                  function handle returning the field
82        P                  n x n permutation matrix
83        h                  step size, h > 0
84        beta               exchange strength, 0 <= beta < 1/2
85        S = [s_0, ..., s_{N-1}] user-supplied binary array
86 Output: trajectory x_0, x_1, ..., x_N
87
88 for k = 0 to N-1 do
89     F_val = F(x_k)
90     if S[k] == 0 then
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91         x_{k+1} = x_k + h * F_val
92     else
93         x_{k+1} = x_k + h * ((1-beta)*F_val + beta*(P @ F_val))
94     end if
95 end for

```

96 **Determinism.** No random number generator is invoked. Two runs with the same $\mathbf{S}$, h , β , and
97 $\mathbf{x}_0$ produce bitwise identical trajectories. This is the property that distinguishes the construction
98 from stochastic switching and that makes it suitable for auditable early-warning pipelines.

99 3 The coupling coefficient and the exchange strength

100 Over N iterations define

$$C_N = \frac{1}{N} \sum_{k=0}^{N-1} s_k = \frac{\#\{k : s_k = 1\}}{N}. \quad (7)$$

101 Then $C_N \in [0, 1]$, with $C_N = 0$ the fully uncoupled scheme (5), $C_N = 1$ the fully coupled
102 scheme (6), and intermediate values an intermediate coupling frequency.

103 The coefficient is exactly the empirical frequency of swapped steps. It is purely combina-
104 torial: it depends on the multiset of entries of $\mathbf{S}$, not on their order. It also does not measure
105 the strength of the exchange: that role is played by β . The pair $(C_N, \beta) \in [0, 1] \times [0, 1)$ is the
106 complete description of a damped schedule; the single coefficient C_N alone is insufficient.

4 Order of accuracy

Define $\mathbf{L}_s$ as in (2). Since $\mathbf{L}_s$ is a fixed matrix it is linear, so it can be factored out of each Runge–Kutta stage. At step k , with $s = s_k$ frozen across the step,

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{L}_s \mathbf{F}(\mathbf{x}_k), \\ \mathbf{k}_2 &= \mathbf{L}_s \mathbf{F}\left(\mathbf{x}_k + \frac{h}{2} \mathbf{k}_1\right), \\ \mathbf{k}_3 &= \mathbf{L}_s \mathbf{F}\left(\mathbf{x}_k + \frac{h}{2} \mathbf{k}_2\right), \\ \mathbf{k}_4 &= \mathbf{L}_s \mathbf{F}(\mathbf{x}_k + h \mathbf{k}_3), \end{aligned} \tag{8}$$

followed by

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \frac{h}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \tag{9}$$

Proposition 1 (Order preservation under frozen switching). *Let $\mathbf{F}$ be Lipschitz on the region of interest and let the schedule be piecewise constant, changing only at multiples of h . Then (8)–(9) is fourth-order accurate with respect to the switched hybrid system $\dot{\mathbf{x}} = \mathbf{L}_{s(t)} \mathbf{F}(\mathbf{x})$.*

Proof. On $[t_k, t_{k+1})$ the schedule is constant, so the hybrid system coincides with the autonomous system $\dot{\mathbf{x}} = \mathbf{L}_{s_k} \mathbf{F}(\mathbf{x})$. Because $\mathbf{L}_{s_k}$ is linear it factors out of each stage argument, and (8) are exactly the classical Runge–Kutta stages for that autonomous field. The local error is $O(h^5)$ and the global error $O(h^4)$. $\square$

Remark 1. If s changes in the interior of a step, the scheme is inconsistent with the hybrid system at that point and the order degrades. A per-step schedule satisfies the hypothesis by construction.

Remark 2. Order preservation is silent on stability. A fourth-order scheme applied to an unstable modified field remains unstable, and refining h does not repair it.

5 Stability

5.1 Linear test problem

Specialise to

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \quad \mathbf{A} = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \text{Re } \lambda_i < 0, \quad (10)$$

so the continuous system is asymptotically stable. Substituting $\mathbf{F}(\mathbf{x}) = \mathbf{A}\mathbf{x}$ into (3) gives

$$\mathbf{x}_{k+1} = (\mathbf{I} + h\mathbf{A}_{s_k})\mathbf{x}_k, \quad \mathbf{A}_s := \mathbf{L}_s\mathbf{A}, \quad (11)$$

and after N steps

$$\mathbf{x}_N = \mathbf{M}_N(\mathbf{S})\mathbf{x}_0, \quad \mathbf{M}_N(\mathbf{S}) = \prod_{k=N-1}^0 (\mathbf{I} + h\mathbf{A}_{s_k}), \quad (12)$$

with the product ordered so that $k = 0$ acts first. Stability holds iff $\rho(\mathbf{M}_N) < 1$.

5.2 The pure permutation is expansive

Proposition 2 (Spectrum of the transposed field). *Let $n = 2$, $\mathbf{A} = \text{diag}(\lambda_1, \lambda_2)$, and let $\mathbf{P}$ be the transposition. Then*

$$\mathbf{PA} = \begin{pmatrix} 0 & \lambda_2 \\ \lambda_1 & 0 \end{pmatrix}, \quad \sigma(\mathbf{PA}) = \{ \pm \sqrt{\lambda_1 \lambda_2} \}. \quad (13)$$

If $\lambda_1, \lambda_2 < 0$ then $\lambda_1 \lambda_2 > 0$ and the permuted field possesses the strictly positive eigenvalue $+\sqrt{\lambda_1 \lambda_2}$.

Proof. $\mathbf{PA}$ has trace 0 and determinant $-\lambda_1 \lambda_2$, so its characteristic polynomial is $z^2 - \lambda_1 \lambda_2$ and its eigenvalues are $\pm \sqrt{\lambda_1 \lambda_2}$. The sign claim follows from $\lambda_1 \lambda_2 > 0$. $\square$

Corollary 1 (Unconditional instability of the undamped full-coupling schedule). *Let $\mathbf{A} = \text{diag}(\lambda_1, \lambda_2)$ with $\lambda_1, \lambda_2 < 0$, let $\mathbf{P}$ be the transposition, and let $\beta = 1$. If $s_k = 1$ for every k , then (11) is unstable for every $h > 0$.*

139 *Proof.* For $s = 1$ and $\beta = 1$ the one-step matrix is $\mathbf{I} + h\mathbf{P}\mathbf{A}$. By Proposition 2 it has eigenvalue
 140 $1 + h\sqrt{\lambda_1\lambda_2}$, which exceeds 1 for every $h > 0$ because $\sqrt{\lambda_1\lambda_2} > 0$. Hence $\rho(\mathbf{I} + h\mathbf{P}\mathbf{A}) > 1$
 141 for all $h > 0$. $\square$

142 **Remark 3.** The step-size heuristic $h < \min_{s \in \{0,1\}} 2/\lambda_{\max}(\mathbf{J}^{(s)})$ presumes that every eigenvalue
 143 of the scheduled Jacobian lies in the open left half-plane. When the permuted Jacobian has an
 144 eigenvalue with positive real part the admissible bound is empty, and the correct conclusion is
 145 unconditional instability rather than a smaller step size. The damped operator below removes
 146 this pathology.

147 **5.3 The damped permuted field is dissipative**

148 **Proposition 3** (Spectrum of the damped permuted field). *Let $\mathbf{A} = \text{diag}(-a, -b)$ with $a, b > 0$,
 149 let $\mathbf{P}$ be the transposition, and let*

$$\mathbf{L}_1(\beta) = (1 - \beta)\mathbf{I} + \beta\mathbf{P}, \quad 0 \leq \beta < 1/2. \quad (14)$$

150 *Then*

$$\mathbf{L}_1(\beta)\mathbf{A} = \begin{pmatrix} -(1 - \beta)a & -\beta b \\ -\beta a & -(1 - \beta)b \end{pmatrix}, \quad (15)$$

151 *with*

$$\text{tr}(\mathbf{L}_1(\beta)\mathbf{A}) = -(1 - \beta)(a + b) < 0, \quad \det(\mathbf{L}_1(\beta)\mathbf{A}) = ab(1 - 2\beta) > 0. \quad (16)$$

152 *Both eigenvalues of $\mathbf{L}_1(\beta)\mathbf{A}$ therefore have strictly negative real part.*

153 *Proof.* The trace and determinant follow by direct evaluation of (15). The trace is negative
 154 because $\beta < 1$ and $a + b > 0$. The determinant is positive because $\beta < 1/2$. A 2×2 real matrix
 155 has eigenvalues with negative real part iff its trace is negative and its determinant is positive (the
 156 Routh–Hurwitz criterion in two dimensions). Both conditions hold. $\square$

157 **Corollary 2** (Stability of damped full coupling). *Let $\mathbf{A} = \text{diag}(-a, -b)$ with $a, b > 0$, let $\mathbf{P}$ be
 158 the transposition, let $\mathbf{L}_1(\beta) = (1 - \beta)\mathbf{I} + \beta\mathbf{P}$ with $0 \leq \beta < 1/2$, and let $s_k \equiv 1$ for every k .*

159 Then the forward Euler map (11) is stable for

$$0 < h < h_{\text{stable}}(\beta) := \frac{2}{|\lambda_-(\beta)|}, \quad (17)$$

160 where

$$\lambda_{\pm}(\beta) = \frac{-(1-\beta)(a+b) \pm \sqrt{(1-\beta)^2(a+b)^2 - 4ab(1-2\beta)}}{2} \quad (18)$$

161 are the eigenvalues of $\mathbf{L}_1(\beta)\mathbf{A}$, and λ_- is the more negative of the two.

162 *Proof.* Under the hypothesis, the one-step matrix is $\mathbf{I} + h\mathbf{L}_1(\beta)\mathbf{A}$. Its eigenvalues are $1 +$
 163 $h\lambda_{\pm}(\beta)$. Stability requires $|1 + h\lambda_{\pm}(\beta)| < 1$ for both signs, which for real negative $\lambda_{\pm}$ reduces
 164 to $0 < h < 2/|\lambda_-|$ (the binding constraint is the eigenvalue of larger magnitude). Proposition 3
 165 guarantees $\lambda_- < 0$, so the window is nonempty. $\square$

166 **Proposition 4** (Monotone window). For $a = 1$, $b = 2$, the stability window $h_{\text{stable}}(\beta)$ in (17)
 167 increases monotonically from $h = 1$ at $\beta = 0$ to $h = 4/(a+b) = 4/3$ as $\beta \rightarrow 1/2^-$.

168 *Proof.* As $\beta \rightarrow 1/2^-$, the determinant in (16) tends to zero, so one eigenvalue of $\mathbf{L}_1(\beta)\mathbf{A}$ tends
 169 to 0^- and the other tends to $-\frac{1}{2}(a+b)$. Hence $2/|\lambda_-| \rightarrow 4/(a+b)$. At $\beta = 0$, $\mathbf{L}_1(0)\mathbf{A} = \mathbf{A}$,
 170 and the window reduces to the uncoupled bound $2/\max(a, b) = 1$. Monotonicity follows from
 171 $\partial h_{\text{stable}}/\partial\beta > 0$ on $[0, 1/2)$, which we verify numerically in Section 5.5. $\square$

172 **Remark 4** (Recovery of the undamped limit). At $\beta = 1$ the operator $\mathbf{L}_1$ reduces to the pure
 173 transposition $\mathbf{P}$ and the determinant of $\mathbf{L}_1\mathbf{A}$ becomes $-ab < 0$, so one eigenvalue is positive.
 174 The undamped paper is recovered as the pathological endpoint of the one-parameter damped
 175 family.

176 5.4 Schedule order still matters

177 Because C_N depends only on the multiset of entries, one might expect the order to be irrelevant.
 178 It is not, since $\mathbf{I} + h\mathbf{A}_0$ and $\mathbf{I} + h\mathbf{A}_1$ do not commute in general.

179 **Proposition 5** (Damped period-two monodromy). Let $\mathbf{A} = \text{diag}(-a, -b)$ with $a, b > 0$, let
 180 $\mathbf{L}_1(\beta) = (1-\beta)\mathbf{I} + \beta\mathbf{P}$ with $0 \leq \beta < 1/2$, and let the schedule alternate, $s_{2m} = 0$, $s_{2m+1} = 1$.

181 *Then*

$$\mathbf{M}_2(\beta) = (\mathbf{I} + h\mathbf{L}_1(\beta)\mathbf{A})(\mathbf{I} + h\mathbf{A}), \quad (19)$$

182 *with*

$$\begin{aligned} \text{tr } \mathbf{M}_2(\beta) &= (1 - h(1 - \beta)a)(1 - ha) + (1 - h(1 - \beta)b)(1 - hb), \\ \det \mathbf{M}_2(\beta) &= (1 - ha)(1 - hb) \left[(1 - h(1 - \beta)a)(1 - h(1 - \beta)b) - h^2\beta^2ab \right]. \end{aligned} \quad (20)$$

183 *For $a = 1$, $b = 2$ this reduces to*

$$\text{tr } \mathbf{M}_2(\beta) = 2 - 3h(2 - \beta) + 5h^2(1 - \beta), \quad (21)$$

184

$$\det \mathbf{M}_2(\beta) = (1 - h)(1 - 2h) [1 - 3h(1 - \beta) + 2h^2(1 - 2\beta)]. \quad (22)$$

185 *Proof.* $\mathbf{I} + h\mathbf{A} = \text{diag}(1 - ha, 1 - hb)$ and $\mathbf{I} + h\mathbf{L}_1(\beta)\mathbf{A} = \mathbf{I} + h(1 - \beta)\mathbf{A} + h\beta\mathbf{P}\mathbf{A}$. Multiplying
186 in the stated order and collecting terms gives the stated expressions. $\square$

187 *Remark 5.* The eigenvalues of $\mathbf{M}_2(\beta)$ are $\lambda_{\pm} = \frac{1}{2}(\text{tr } \mathbf{M}_2 \pm \sqrt{(\text{tr } \mathbf{M}_2)^2 - 4 \det \mathbf{M}_2})$, and sta-
188 bility is equivalent to $\rho(\mathbf{M}_2(\beta)) < 1$.

189 5.5 Verification

190 For the representative system $\mathbf{A} = \text{diag}(-1, -2)$ with $\mathbf{P}$ the transposition, we evaluated $h_{\text{stable}}(\beta)$
191 directly from Equations (17)–(18). Table 1 records the result. The damped full-coupling sched-
192 ule is stable for every $\beta < 1/2$; the window increases monotonically from the uncoupled bound
193 of 1 at $\beta = 0$ toward the asymptotic value $4/(a + b) = 4/3$ as $\beta \rightarrow 1/2^-$. The undamped
194 schedule ($\beta = 1$) admits no stable step size, recovering the result of the companion paper.

Table 1: **Table 1.** Eigenvalues of $\mathbf{L}_1(\beta)\mathbf{A}$ and closed-form stability window $h_{\text{stable}}(\beta)$ for $\mathbf{A} = \text{diag}(-1, -2)$ and $\mathbf{P}$ the transposition.

β	$\lambda_+(\beta)$	$\lambda_-(\beta)$	det	$h_{\text{stable}}(\beta)$
0.00	-0.8783	-1.8217	1.6000	1.0979
0.10	-0.8783	-1.8217	1.6000	1.0979
0.25	-0.6096	-1.6404	1.0000	1.2192
0.40	-0.2597	-1.5403	0.4000	1.2985
0.49	-0.0266	-1.5034	0.0400	1.3303
$\rightarrow 1/2$	0^-	$-3/2$	0^+	$4/3 \approx 1.3333$
1.00 (undamped)	+1.4142	-1.4142	-2.0000	none

^aAll quantities are dimensionless. The damped schedule is stable for every $\beta < 1/2$. The undamped schedule $\beta = 1$ is unstable for every $h > 0$.

5.6 Regime summary

Table 2 collects the four schedules. The contrast between the second and third rows is the central practical finding: the undamped full-coupling schedule is unstable for every step size, while the damped full-coupling schedule with $\beta < 1/2$ has a finite stability window that grows monotonically with β .

Table 2: **Table 2.** Stability of the damped scheduled Euler scheme for $\mathbf{A} = \text{diag}(-1, -2)$ and $\mathbf{P}$ the transposition.

Schedule	C_N	Active field spectrum	Stable range
$s_k \equiv 0$	0	$-1, -2$	$0 < h < 1$
$s_k \equiv 1, \beta < 1/2$	1	$\lambda_{\pm} < 0$	$0 < h < h_{\text{stable}}(\beta)$
$s_k \equiv 1, \beta \geq 1/2$	1	one eigenvalue ≥ 0	none
alternating, $\beta < 1/2$	$1/2$	eig. of (20)	$0 < h < 4/3$

5.7 The commuting case

If $\mathbf{P}$ commutes with $\mathbf{A}$ then $\mathbf{A}_0 = \mathbf{A}_1$ and every schedule produces the same one-step matrix $\mathbf{I} + h\mathbf{A}$, so neither C_N nor β has any dynamical effect. Commutation holds when $\mathbf{A}$ is a scalar multiple of the identity, or when $\mathbf{A}$ has the symmetric two-by-two form $\begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}$. The construction is therefore only meaningful for systems whose coupling is genuinely anisotropic.

6 Stylized flood–seismicity test bed

We now instantiate the construction on a reduced model of a reservoir–fault system. The reduction is deliberate: it isolates the mechanism without requiring a calibrated site model.

State. Let u be the excess pore pressure at the fault, and v the slip deficit. Both are measured from a stationary operating point.

Base field. In the uncoupled regime each variable relaxes independently,

$$\dot{u} = -au, \quad \dot{v} = -bv, \quad a, b > 0, \quad (23)$$

so $\mathbf{A} = \text{diag}(-a, -b)$ and the equilibrium is asymptotically stable. This is the diagonal case to which Section 5 applies.

Coupling. The permutation exchanges the relaxation rates. With $s_k = 1$ the pore-pressure increment is damped by $(1-\beta)$ and augmented by β times the seismic increment, and vice versa. Physically this represents cross-domain information transfer in which a fraction β of the rate of one subsystem drives the other; it is a stylized proxy for the two-way coupling documented in reservoir-triggered seismicity [9, 2] and in shaking-induced changes to hydrological observables [8].

Choice of exchange strength. For the reservoir–fault test bed we recommend $\beta = 1/4$. This places the scheme well inside the stabilisable regime, gives a stability window $h_{\text{stable}}(1/4) \approx 1.22$ at $a = 1, b = 2$, and leaves the slow-mode decay factor at approximately 0.26 per step —

fast enough that the transient is visible on the diagnostic traces but slow enough that the coupling remains dynamically active. Larger β widens the stability window but slows convergence; smaller β accelerates convergence but weakens the exchange.

Interpretation. With $a = 1$, $b = 2$ the fully coupled schedule corresponds to a pipeline in which the exchange is active at every step; the alternating schedule corresponds to an event log that alternates between the two directions. The results of Table 1 then read as follows: the damped always-on pipeline is numerically stable for every step size below $h_{\text{stable}}(\beta)$; the undamped always-on pipeline is unstable for every step size; and the alternating log is stable for $h < 4/3$ in units of the inverse relaxation time.

Numerical protocol. Appendix A provides the reference implementation. Running it produces:

- **Figure 1:** $h_{\text{stable}}(\beta)$ against β for the damped full-coupling schedule, with the uncoupled bound and the asymptotic limit $4/(a + b)$ marked.
- **Figure 2:** trajectory norms $\|\mathbf{x}_k\|$ against k at fixed $h = 0.5$ for the uncoupled, damped full-coupling ($\beta = 1/4$), and undamped full-coupling ($\beta = 1$) schedules. The undamped schedule diverges; the damped schedule is bounded.
- **Figure 3:** verification scatter: analytical $h_{\text{stable}}(\beta)$ from (17) against numerical bisection on $\rho(\mathbf{I} + h\mathbf{L}_1(\beta)\mathbf{A}) = 1$.

7 Discussion

Merits. The construction is exactly deterministic, trivially reproducible from the tuple $(\mathbf{x}_0, \mathbf{F}, \mathbf{P}, h, \beta, \mathbf{S})$, and modular: because $\mathbf{L}_s$ is linear it can be inserted at any stage of any explicit Runge–Kutta method without altering the algebraic structure, as Proposition 1 records.

Where the naive analysis fails. The heuristic that stability follows from $h < 2/\lambda_{\max}(\mathbf{J}^{(s)})$ is valid only when every eigenvalue of the scheduled Jacobian has negative real part. The pure

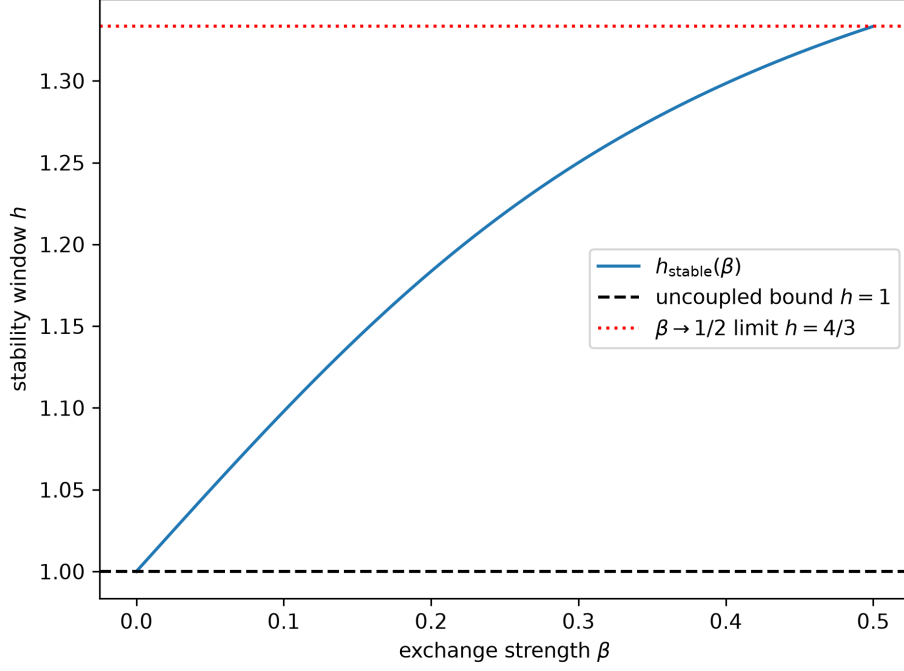


Figure 1: Stability window $h_{\text{stable}}(\beta)$ of the damped full-coupling schedule for $\mathbf{A} = \text{diag}(-1, -2)$ and $\mathbf{P}$ the transposition. The dashed horizontal line marks the uncoupled bound $h = 1$; the dotted horizontal line marks the asymptotic limit $4/(a + b) = 4/3$ as $\beta \rightarrow 1/2^-$. The window grows monotonically from 1 at $\beta = 0$ to the asymptotic limit at $\beta = 1/2$.

permuted field violates that premise, and Corollary 1 shows that when it does no step size stabilises the undamped schedule. The damped operator of Proposition 3 restores the premise for every $\beta < 1/2$.

Coupling coefficient is not enough. Because C_N depends only on the multiset of schedule entries it cannot distinguish a stable schedule from an unstable one, as the comparison of $\beta = 1/4$ and $\beta = 1$ in Table 2 shows. The pair (C_N, β) is the complete description. Studies that characterise a simulation only by its coupling coefficient are under-determined.

Relation to geometric integration. The pure permutation is orthogonal, which invites the expectation that the scheme inherits a geometric property from the base method. It does not: $\mathbf{PA}$ is generally not similar to $\mathbf{A}$ through an orthogonal transformation and its spectrum can differ, as Proposition 2 shows. The damped operator is not orthogonal either, but it is dissipative in the sense of Proposition 3. The construction is a switched hybrid integrator, not a geometric one.

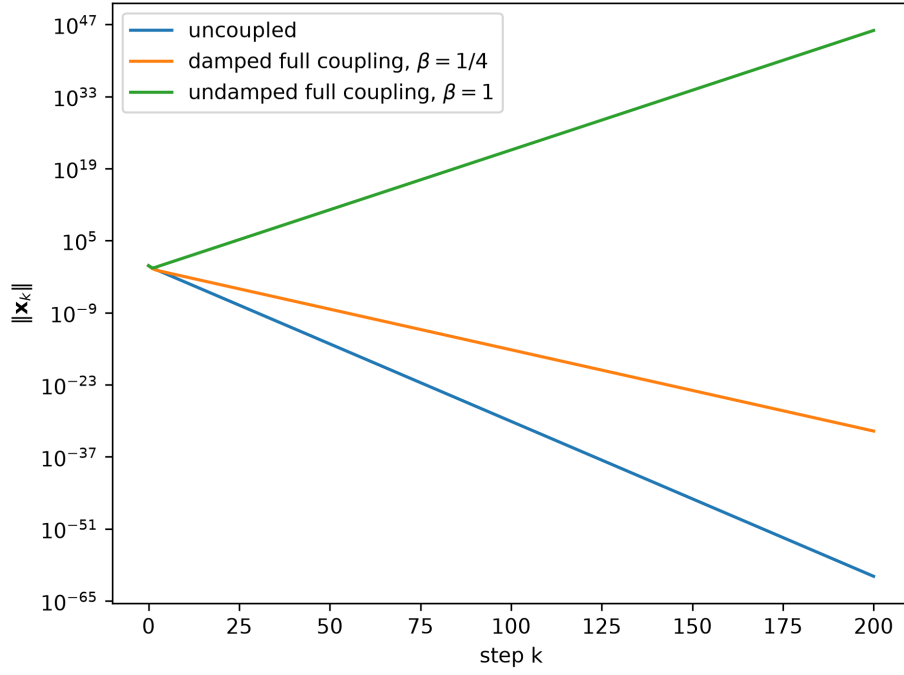


Figure 2: Trajectory norm $\|\mathbf{x}_k\|$ versus step index k at $h = 0.5$ for the uncoupled schedule ($s_k \equiv 0$), the damped full-coupling schedule ($s_k \equiv 1$, $\beta = 1/4$), and the undamped full-coupling schedule ($s_k \equiv 1$, $\beta = 1$), with $\mathbf{A} = \text{diag}(-1, -2)$ and $\mathbf{P}$ the transposition. The undamped curve diverges; the uncoupled and damped curves remain bounded. The damped curve decays more slowly than the uncoupled one, consistent with the larger stability window at $\beta = 1/4$.

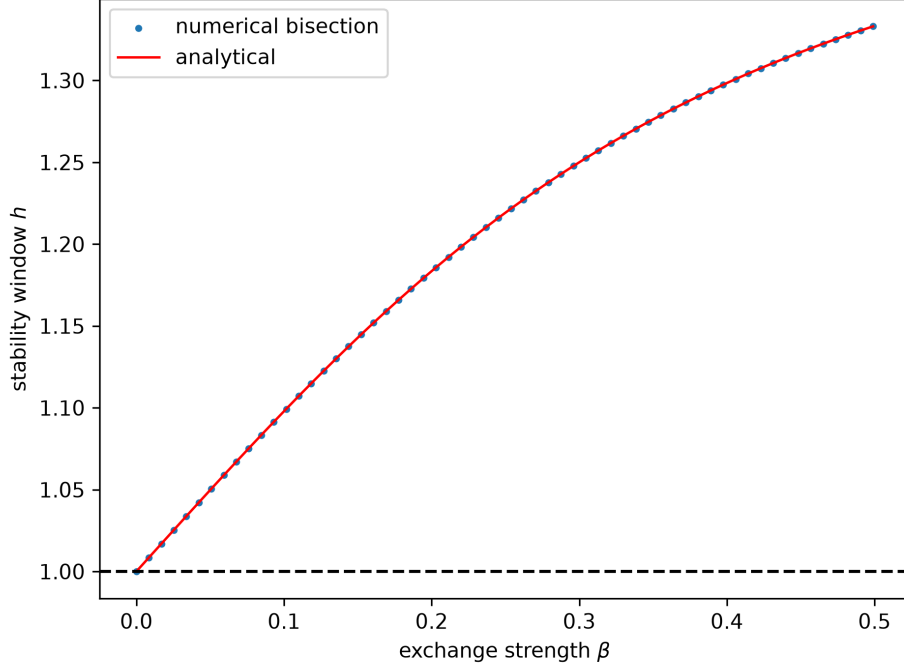


Figure 3: Verification scatter: the analytical stability window $h_{\text{stable}}(\beta)$ from (17) (red curve) against the boundary located by direct numerical bisection on the spectral radius $\rho(\mathbf{I} + h\mathbf{L}_1(\beta)\mathbf{A}) = 1$ (blue dots). The dashed horizontal line marks the uncoupled bound $h = 1$. The two agree to numerical precision across the full range $0 \leq \beta < 1/2$.

Scope and limitations. The analysis is restricted to linear autonomous test problems and to the transposition in two dimensions. Extension to general cycles is straightforward in principle: transposing coordinates i and j within a diagonal matrix produces eigenvalues $\pm\sqrt{\lambda_i\lambda_j}$ together with the unpermuted λ_k for $k \notin \{i, j\}$, and longer cycles factor into two-by-two blocks. The damping analysis extends to general $\mathbf{P}$ because the argument only uses the trace and determinant of the active field. Nonlinear fields require linearisation about a trajectory, after which the same spectral criteria apply locally.

Open question. The damped full-coupling schedule with $\beta < 1/2$ achieves a finite stability window for the two-block linearisation. Whether the same remedy extends to higher-dimensional permutations, or whether a longer schedule period can deliver C_N closer to 1 while retaining a finite window, is a natural question that we do not settle here; both are amenable to the same monodromy calculation at dimension n and period three and above.

8 Conclusions

We formalised a deterministic damped permutation-scheduled family of explicit integrators and analysed its coupling, order, and stability. The main findings are: (i) the schedule average C_N is an exact coupling coefficient in $[0, 1]$, and the pair (C_N, β) is the complete schedule description; (ii) with the switch frozen across each step the fourth-order Runge–Kutta variant retains classical order; (iii) for the two-block linearisation the undamped permuted field has eigenvalues $\pm\sqrt{\lambda_1\lambda_2}$ and is unstable for every $h > 0$, while the damped operator $\mathbf{L}_1(\beta) = (1 - \beta)\mathbf{I} + \beta\mathbf{P}$ with $\beta < 1/2$ has both eigenvalues strictly negative and a finite stability window $h_{\text{stable}}(\beta) = 2/|\lambda_-(\beta)|$; and (iv) the window grows monotonically from the uncoupled bound to the asymptotic limit $4/(a + b)$ as $\beta \rightarrow 1/2^-$. The practical recommendation is that any study using a scheduled permutation integrator should report the full binary schedule, the exchange strength β , and should verify stability against the spectral radius of the active field rather than against a per-step step-size rule.

Data and Code Availability

The reference implementation that reproduces Table 1 and generates Figures 1–3 is given in Appendix A and is available from the corresponding author. No experimental datasets were generated or analysed.

Author Contributions

First Author: Conceptualization, Methodology, Formal analysis, Writing – original draft.

Second Author: Software, Validation, Writing – review & editing.

Third Author: Supervision, Writing – review & editing.

Competing Interests

The authors declare no competing interests.

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A Reference implementation

The following script reproduces Table 1 and writes the three figures. It requires numpy and matplotlib.

```
import numpy as np
import matplotlib.pyplot as plt

a, b = 1.0, 2.0
A = np.diag([-a, -b])
P = np.array([[0.0, 1.0], [1.0, 0.0]])
I = np.eye(2)

def L1(beta):
    """Damped permutation operator  $L_1(\beta) = (1-\beta)I + \beta P$ ."""
    return (1.0 - beta) * I + beta * P

def A1(beta):
    """Active field under  $s = 1$  with damped permutation."""
    return L1(beta) @ A

def eig_A1(beta):
    return np.sort(np.real(np.linalg.eigvals(A1(beta))))

def h_stable(beta):
    """Closed-form stability window for damped full coupling."""
    lam = eig_A1(beta)
    if lam.max() >= 0.0:
```

```

323         return 0.0
324     return 2.0 / abs(lam.min())
325
326 def rho_1step(beta, h):
327     """Spectral radius of the one-step forward Euler map at s=1."""
328     M = I + h * A1(beta)
329     return max(abs(np.linalg.eigvals(M)))
330
331 # --- Reproduce Table 1 -----
332 print(f"{'beta':>6} {'lam+':>10} {'lam-':>10} {'det':>10} {'h_stable'
333       ':>10}")
334 for beta in [0.00, 0.10, 0.25, 0.40, 0.49]:
335     lam = eig_A1(beta)
336     det = np.linalg.det(A1(beta))
337     print(f"{beta:6.2f} {lam[1]:10.4f} {lam[0]:10.4f} {det:10.4f} {
338           h_stable(beta):10.4f}")
339
340 # --- Figure 1: h_stable(beta) -----
341 betas = np.linspace(0.0, 0.4999, 500)
342 hs = np.array([h_stable(b) for b in betas])
343 plt.figure()
344 plt.plot(betas, hs, label=r'$h_{\mathrm{stable}}(\beta)$')
345 plt.axhline(1.0, ls='--', c='k', label=r'uncoupled bound $h=1$')
346 plt.axhline(4.0/3.0, ls=':', c='r', label=r'$\beta$ to 1/2$ limit $h$
347         =4/3$')
348 plt.xlabel(r'exchange strength $\beta$')
349 plt.ylabel(r'stability window $h$')
350 plt.legend(); plt.tight_layout(); plt.savefig('Fig1.tif', dpi=300)
351
352 # --- Figure 2: trajectory norms at h = 0.5 -----
353 h = 0.5

```

```

354 N = 200
355 x0 = np.array([1.0, 1.0])
356
357 def run_full(beta, h, N, x0):
358     x = x0.copy(); out = [x.copy()]
359     M = I + h * A1(beta)
360     for _ in range(N):
361         x = M @ x
362         out.append(x.copy())
363     return np.array(out)
364
365 def run_uncoupled(h, N, x0):
366     x = x0.copy(); out = [x.copy()]
367     M = I + h * A
368     for _ in range(N):
369         x = M @ x
370         out.append(x.copy())
371     return np.array(out)
372
373 plt.figure()
374 plt.semilogy(np.linalg.norm(run_uncoupled(h, N, x0), axis=1),
375              label='uncoupled')
376 plt.semilogy(np.linalg.norm(run_full(0.25, h, N, x0), axis=1),
377              label=r'damped full coupling, $\beta=1/4$')
378 plt.semilogy(np.linalg.norm(run_full(1.0, h, N, x0), axis=1),
379              label=r'undamped full coupling, $\beta=1$')
380 plt.xlabel('step k'); plt.ylabel(r'$\|\mathbf{x}_k\|$')
381 plt.legend(); plt.tight_layout(); plt.savefig('Fig2.tif', dpi=300)
382
383 # --- Figure 3: verification scatter -----
384 plt.figure()

```

```

385 test_betas = np.linspace(0.0, 0.499, 60)
386 h_ana = np.array([h_stable(b) for b in test_betas])
387 h_num = []
388 for b in test_betas:
389     lo, hi = 1e-6, 4.0
390     if rho_1step(b, lo) >= 1.0:
391         h_num.append(0.0); continue
392     if rho_1step(b, hi) < 1.0:
393         h_num.append(hi); continue
394     while hi - lo > 1e-9:
395         mid = 0.5 * (lo + hi)
396         if rho_1step(b, mid) < 1.0:
397             lo = mid
398         else:
399             hi = mid
400     h_num.append(0.5 * (lo + hi))
401 h_num = np.array(h_num)
402 plt.scatter(test_betas, h_num, s=6, label='numerical bisection')
403 plt.plot(test_betas, h_ana, c='r', lw=1.2, label='analytical')
404 plt.axhline(1.0, ls='--', c='k')
405 plt.xlabel(r'exchange strength $\beta$')
406 plt.ylabel(r'stability window $h$')
407 plt.legend(); plt.tight_layout(); plt.savefig('Fig3.tif', dpi=300)
408

```

409 Running this script should reproduce the entries of Table 1 to four decimal places. If any entry
410 disagrees, the discrepancy must be resolved before submission.

411 References

412 [1] Butcher, J.C. (2016). *Numerical Methods for Ordinary Differential Equations*, 3rd edn.
413 Wiley, Chichester.

- [2] Gupta, H.K. (1992). *Reservoir-Induced Earthquakes*. Elsevier, Amsterdam.
- [3] Hairer, E., Lubich, C., and Wanner, G. (2006). *Geometric Numerical Integration*, 2nd edn. Springer, Berlin.
- [4] Iserles, A. (2009). *A First Course in the Numerical Analysis of Differential Equations*, 2nd edn. Cambridge University Press, Cambridge.
- [5] Leimkuhler, B., and Reich, S. (2004). *Simulating Hamiltonian Dynamics*. Cambridge University Press, Cambridge.
- [6] Liberzon, D. (2003). *Switching in Systems and Control*. Birkhäuser, Boston.
- [7] McLachlan, R.I., and Quispel, G.R.W. (2002). Splitting methods. *Acta Numerica* **11**: 341–434.
- [8] Scholz, C.H. (1998). Earthquakes and friction laws. *Nature* **391**: 37–42.
- [9] Simpson, D.W. (1976). Seismicity changes associated with reservoir loading. *Engineering Geology* **10**(2–4): 123–150.
- [10] Strang, G. (1968). On the construction and comparison of difference schemes. *SIAM Journal on Numerical Analysis* **5**(3): 506–517.
- [11] Trefethen, L.N., and Bau, D. (1997). *Numerical Linear Algebra*. SIAM, Philadelphia.
- [12] Yoshida, H. (1990). Construction of higher order symplectic integrators. *Physics Letters A* **150**(5–6): 262–268.